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RESEARCH PROJECT REPORT ON Static Structural Analysis

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Chapter 1

Introduction

ANSYS Workbench is a powerful simulation platform widely used in engineering fields. It allows students to simulate, analyze, and solve various engineering problems, including structural, thermal, fluid, and electromagnetic applications. This report details the step-by-step process of performing a static structural analysis for a lifting lug using ANSYS Workbench 2023 R1.

1.1 Digital Twin Technology

Digital twins are virtual representations of physical assets or systems, enabling real-time analysis, predictive maintenance, and optimization. Key features of ANSYS digital twins include:

- Simulation-based design and real-time data integration
- IoT compatibility and predictive analytics
- Applications in manufacturing, healthcare, aerospace, and more

For this project, the digital twin concept was applied to predict failures and optimize the lifting lug's performance under load.

1.2 Objective

To assess the structural integrity of a lifting lug, a comprehensive simulation and analysis will be conducted under various applied loads and stresses. This analysis will involve evaluating the Equivalent (von Mises), Maximum Principal, Middle Principal, Minimum Principal, Maximum Shear, Intensity, Normal, Shear, Vector Principal, Error, Membrane, and Bending stresses. By identifying potential areas of stress concentration or failure, this analysis will provide valuable insights into the lug's performance under different loading conditions.

Chapter 2

Step-by-Step Approach

2.1 Methodology

The analysis combined numerical (FEA) and analytical approaches:

- **Numerical (FEA):** ANSYS Workbench 2023 R1 with outputs including von Mises stress, principal stresses, and shear stresses
- **Analytical:** EN 1993-1-8 standards for steel structures

2.2 Step 1: Installing ANSYS Workbench

- Open ANSYS Workbench 2023 R1 by double-clicking its icon
- The workbench will appear with an empty workspace

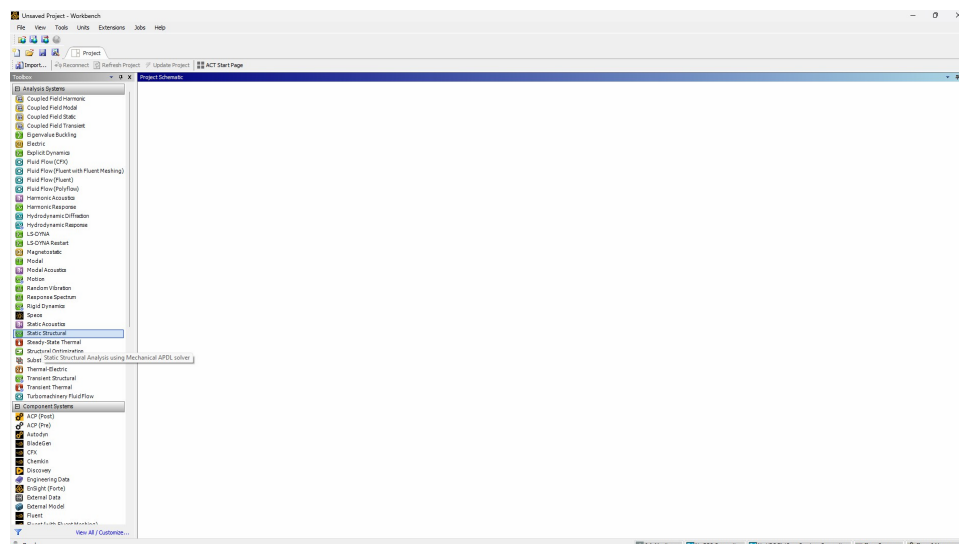


Figure 2.1: Opening ANSYS Workbench

2.3 Step 2: Units of Measurement

- Locate the *Units* option in the toolbar and left-click on it

- Select *Metric (tonne, mm, s, °C, mA, N, mV)* for this project

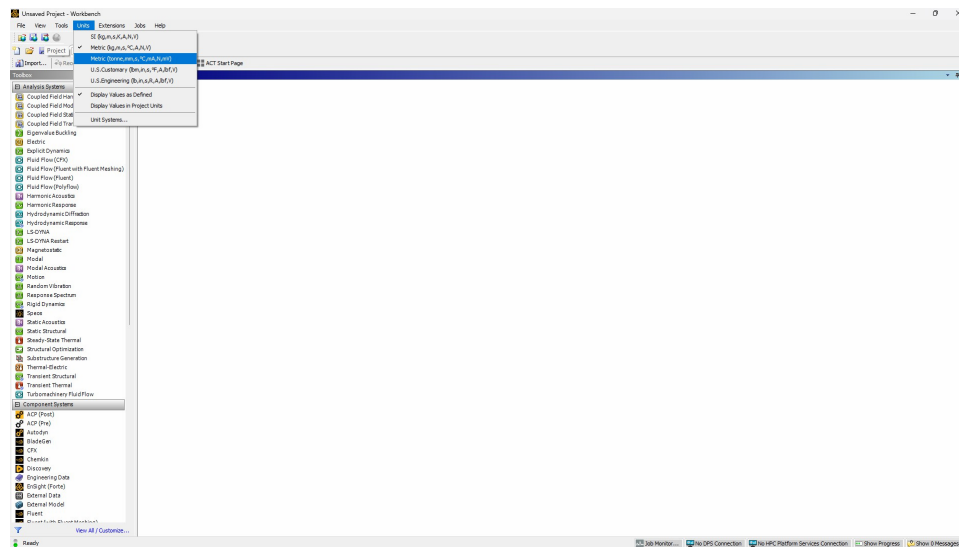


Figure 2.2: Setting Units of Measurement

2.4 Step 3: Identify type of analysis

- To perform static structural analysis, locate it in the toolbox under 'Analysis System'
- Double-click on 'Static Structural' or drag and drop it onto the project schematic

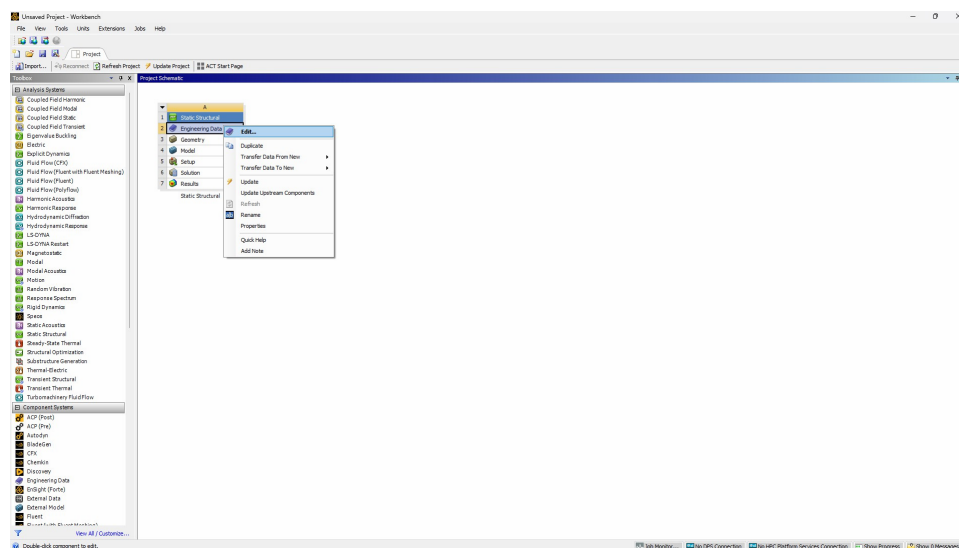


Figure 2.3: Selection of Static Structural

2.5 Step 4: Setup Analysis

- Right-click on 'Engineering Data' and select 'Edit'

- Double-click on 'General Materials' and click on the '+' icon to add the material to the analysis system

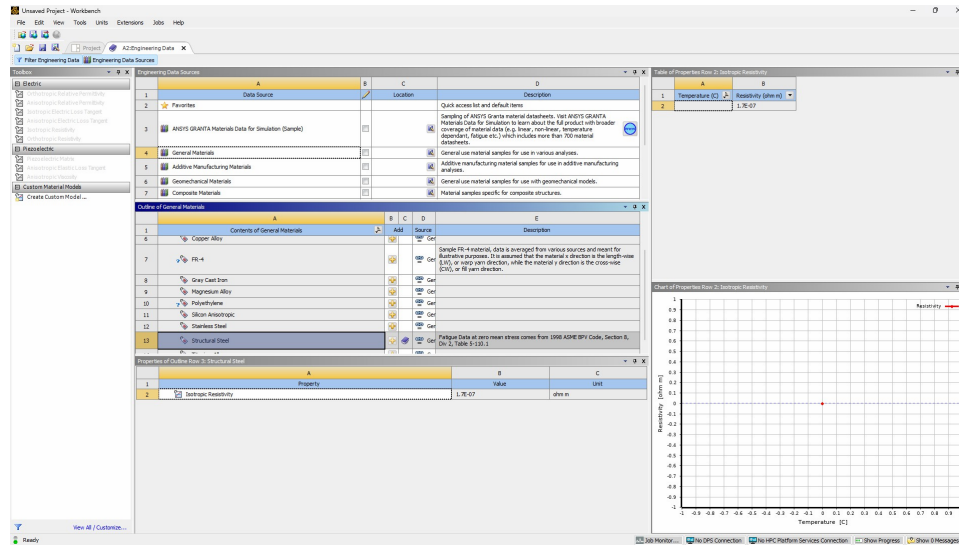


Figure 2.4: Material Assignment

2.6 Step 5: Create Geometry

- Right-click and choose "New SpaceClaim Geometry" to open a design space
- Use the "Rectangle" tool to create a base with desired dimensions
- Extrude the base and switch to sketch mode on a perpendicular surface
- Draw precise features (lines, circles) and extrude selected faces to create the lifting lug shape
- Add a separate plate using sketch and pull functions, ensuring it doesn't merge with the lug

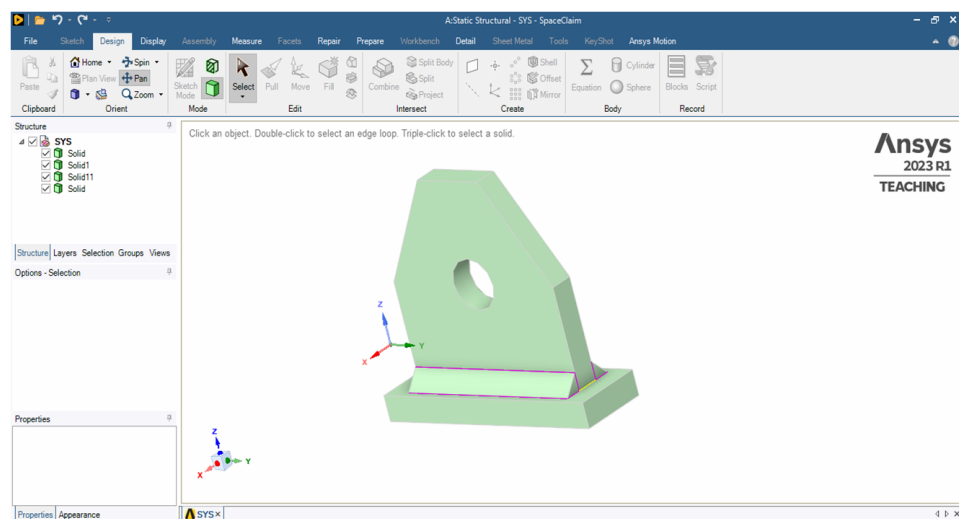


Figure 2.5: Geometry

2.7 Step 6: Model Setup and Connection Setup

- Right-click "Model" in the system box, then select "Edit" and choose the geometry under "Project Tree" for editing
- Assign material by right-clicking the solid, selecting "Details", expanding options, and choosing "Structural Steel"
- Repeat step 2 for each component, then right-click "Connections" and choose "Suppress" to hide them

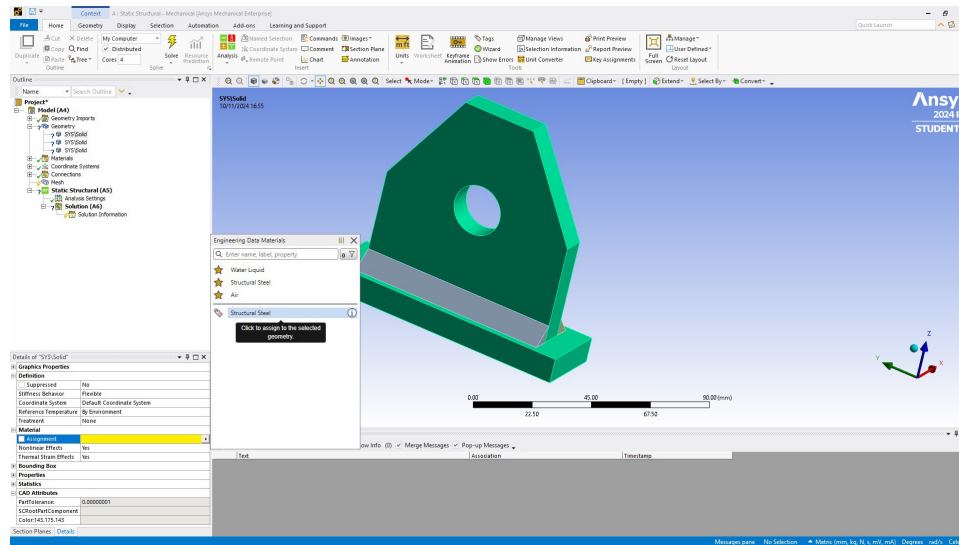


Figure 2.6: Model Setup

2.8 Step 7: Meshing

- Right-click "Mesh" in the Project Tree and select "Generate Mesh"
- Performed element type conversion:
 - **Initial mesh:** Tetrahedral elements (automatic generation)
 - **Optimized mesh:** Hexa elements (structured sweep method (multizone))

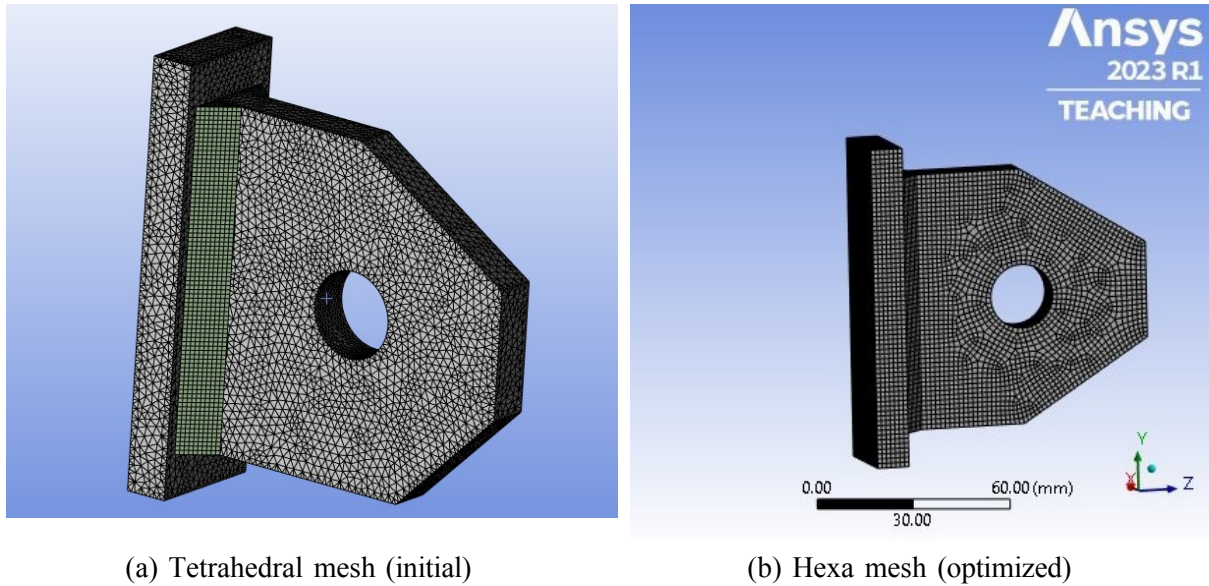


Figure 2.7: Mesh optimization through element conversion

2.8.1 Mesh Optimization Benefits

The transition from tetrahedral to hexa elements provided several benefits, including a noticeable reduction in solution error, leading to improved accuracy. Solver performance also improved with faster convergence. Also the quality of the elements was enhanced, with the aspect ratio improving and skewness decreasing significantly.

2.9 Step 8: Boundary Condition setup

- Right-click "Static Structural" and select "Insert" -> "Remote Force"
- Click "No Selection" beside "Geometry" and select the hole in the lifting lug
- Set the force vector components: $X = 0 \text{ N}$, $Y = 21200 \text{ N}$, $Z = 21200 \text{ N}$
- Repeat steps 1 and 2, select "Fixed Support", and choose the plate surface as the support location
- Click "Apply" to confirm the boundary conditions

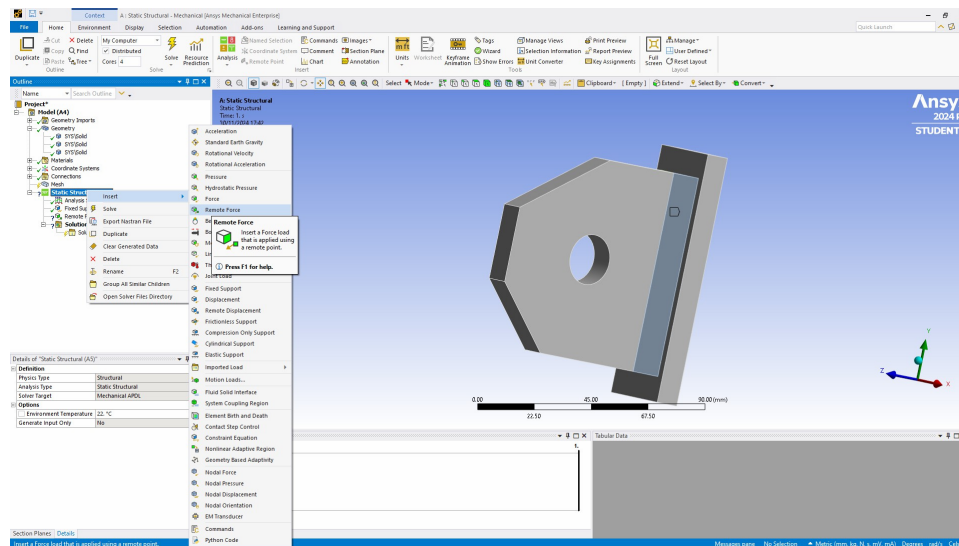


Figure 2.8: Applying Remote Force

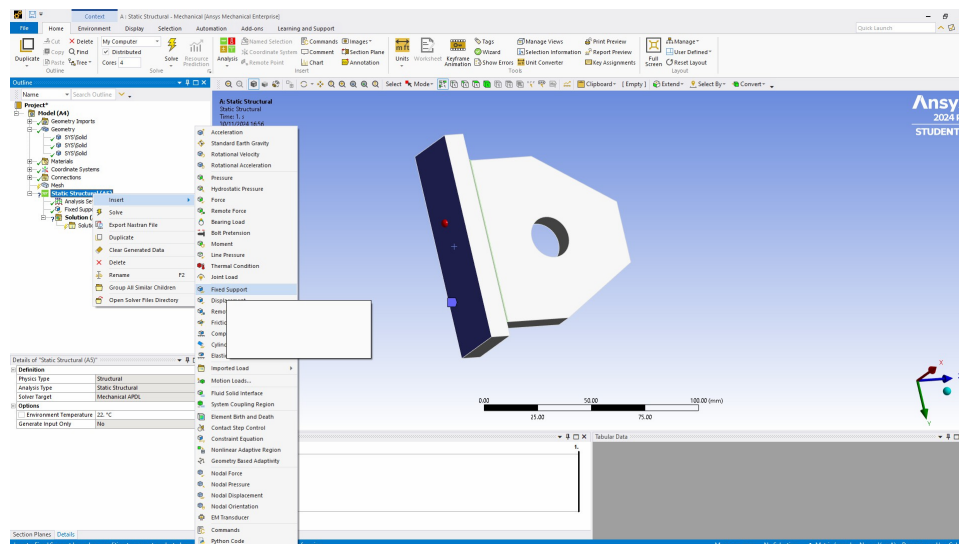


Figure 2.9: Fixed Support

2.10 Step 9: Friction Application Setup

- Expand the "Connections" branch under the Static Structural analysis.
- Right-click "Connections" and select "Insert" –> "Manual Contact Region."
- For "Contact," select the face of the lifting lug that touches the plate.
- For "Target," select the corresponding face on the plate.
- Set "Type" to "Frictional."
- Set the "Friction Coefficient" to 0.2.
- Keep the "Scope Mode" as "Manual" and the other advanced settings as "Program Controlled."

- Click "Apply" to confirm the frictional contact definition.

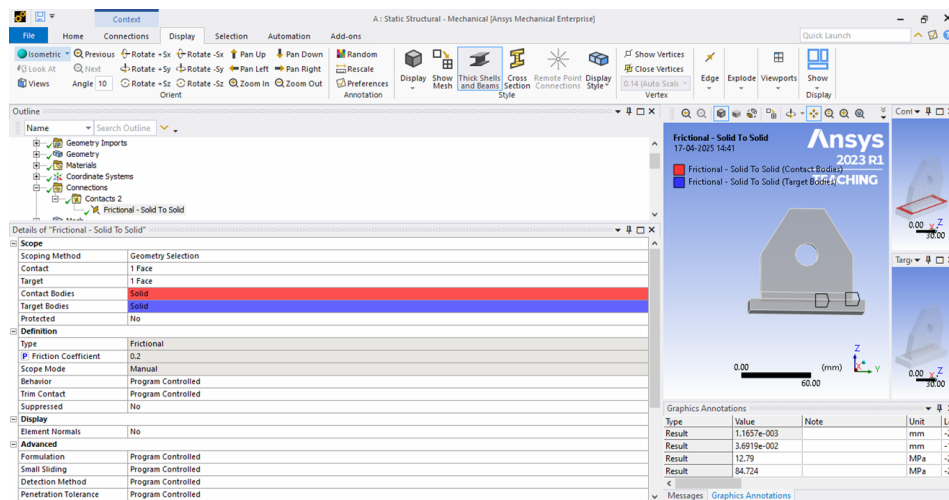


Figure 2.10: Frictional Contact Setup

2.11 Step 10: Add Solution

- Right-click "Solution" in the Project Tree, select "Insert" > "Stress", and choose the desired result
- Repeat step 1 for additional results, then click "Solve" under the "Home" tab to initiate the simulation

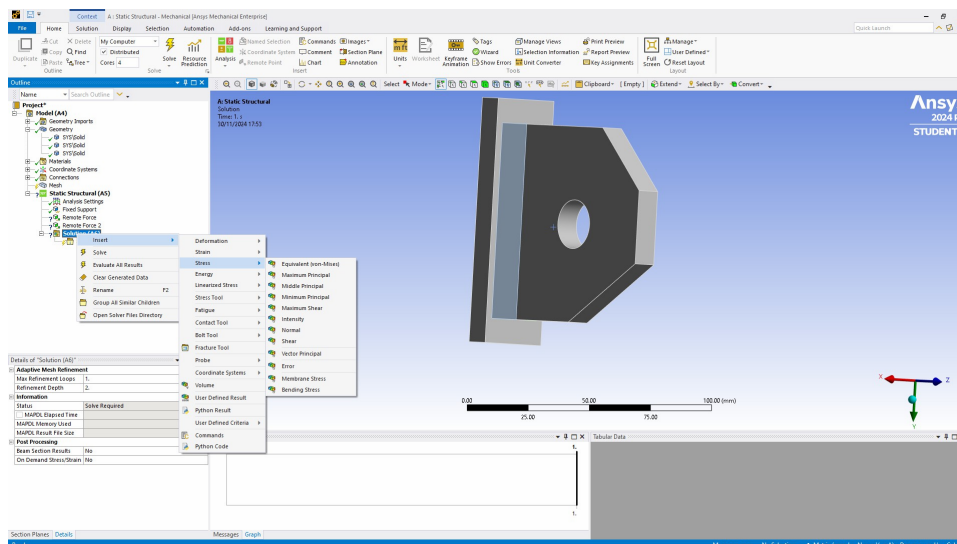


Figure 2.11: Selection of Solutions

Chapter 3

Results and Discussion

3.1 Numerical Analysis (FEA)

The Finite Element Analysis was performed using ANSYS Workbench with different mesh sizes (1.5 mm to 5.0 mm) to investigate mesh sensitivity.

3.1.1 Equivalent von Mises stress

The equivalent von Mises stress is used to predict yielding of materials under complex loading conditions. It provides a single scalar value that can be compared against the material yield strength to assess failure.

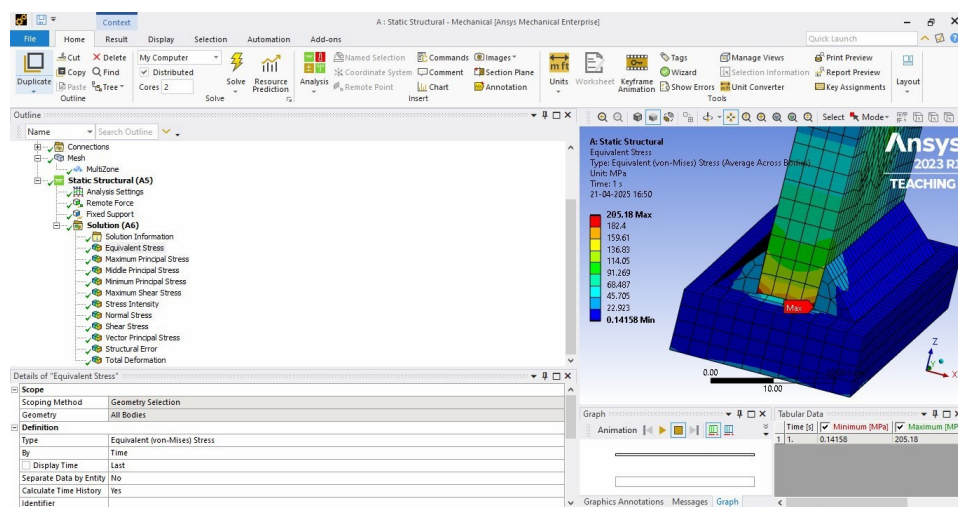
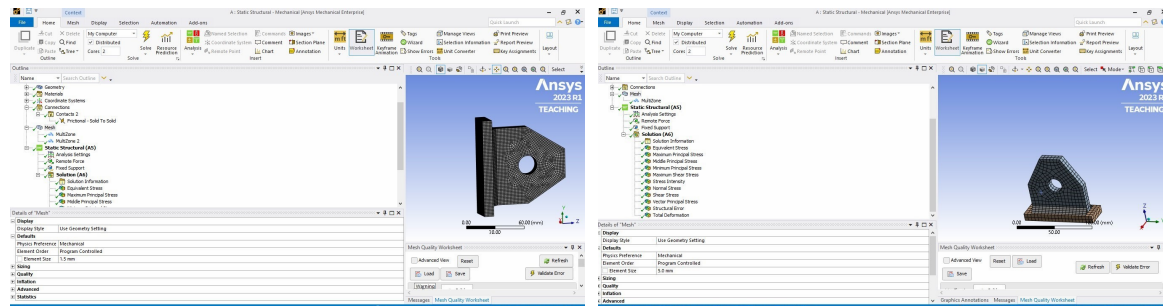


Figure 3.1: Equivalent von Mises stress distribution

3.1.2 Mesh Convergence Analysis

The study evaluated mesh sensitivity:

- **Fine Mesh:** 1.5 mm (critical zones)
- **Coarse Mesh:** 5.0 mm (non-critical zones)



(a) 1.5 mm mesh

(b) 5.0 mm mesh

Figure 3.2: mesh sizes

Key Observations

The fine mesh (1.5 mm) captured localized stress concentrations more precisely but required significantly longer computation time. In contrast, the coarse mesh (5.0 mm) reduced solution time by 74% while maintaining acceptable accuracy (error < 2% compared to analytical results). Overall, for this geometry, a mesh size of 5.0 mm provided the best trade-off between accuracy and computational cost.

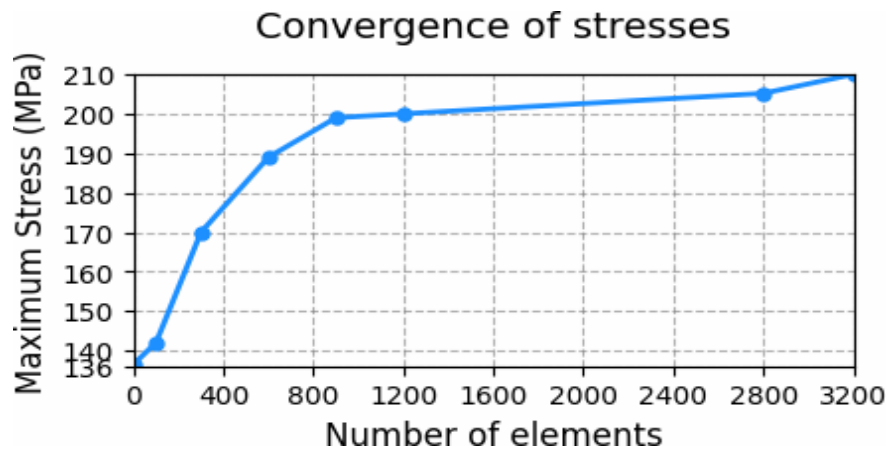


Figure 3.3: Convergence of stresses

3.2 Analytical Method

Based on EN 1993-1-8, the directional method is used for T-joint fillet weld analysis.

3.2.1 Design Equations

$$\sigma_{eq} \leq$$

$$\leq 0.9$$

3.2.2 Input Parameters

- Applied force: $F = 30 \text{ kN}$ at 45°
- Weld throat thickness: $a = 7 \text{ mm}$
- Weld effective length: $L = 80 \text{ mm}$
- Material: S235 steel ($f_u = 360 \text{ MPa}$)
- $\beta_w = 0.8$, $\gamma_{M2} = 1.25$

3.2.3 Step-by-Step Calculation

1. Force Resolution

$$F_x = F \cos(45^\circ) = 30 \times 0.707 = 21.21 \text{ kN}$$

$$F_y = F \sin(45^\circ) = 30 \times 0.707 = 21.21 \text{ kN}$$

2. Throat Area Calculation

$$A_{\text{throat}} = \frac{1}{2} \times a \times L = \frac{1}{2} \times 7 \times 80 = 280 \text{ mm}^2$$

3. Stress Components

- Normal stress ($\sigma_\perp$):

$$\sigma_\perp = \frac{F_x + F_y}{A_{\text{throat}}} \times 10^3 = \frac{42.42}{280} \times 10^3 = 151.5 \text{ MPa}$$

- Shear stress ($\tau_\perp$):

$$\tau_\perp = \frac{F_x}{A_{\text{throat}}} \times 10^3 = \frac{21.21}{280} \times 10^3 = 75.75 \text{ MPa}$$

4. Equivalent Stress Calculation

$$\sigma_{eq} = \sqrt{\sigma_\perp^2 + 3\tau_\perp^2} = 200.4 \text{ MPa}$$

5. Allowable Stresses

$$\frac{f_u}{\gamma_{M2}} = \frac{360}{1.25} = 360 \text{ MPa}$$

$$\beta_w \gamma_{M2} = 0.8 \times 1.25$$

$$\frac{f_u}{\beta_w \gamma_{M2}} = \frac{360}{0.8 \times 1.25} = 0.9 \times 360 = 324 \text{ MPa}$$

= 259.2 MPa

γ_{M2} 1.25

3.2.4 Verification Results

Table 3.1: Analytical Stress Calculations (Updated Geometry)

	Parameter	Value
	Normal stress ($\sigma_{\perp}$)	
		151.5 MPa
Shear stress (τ)		75.75 MPa
Equivalent von Mises stress (σ_{eq})		200.4 MPa
Allowable equivalent stress		360 MPa
Safety factor		1.80

3.2.5 6. Conclusion

Since

$$\sigma_{eq} = 200.4 \text{ MPa} < 360 \text{ MPa} \quad \text{and} \quad \sigma = 151.5 \text{ MPa} < 259.2 \text{ MPa}$$

the weld resistance is sufficient and safe according to the standard.

3.3 Comparative Analysis

Table 3.2: Comparison of FEA and Analytical Results (Updated)

Parameter	FEA	Analytical	Error (%)
Equivalent Stress (MPa)	205.18	200.4	2.33
Max Shear Stress (MPa)	108.13	75.75	42.75
Solution Time (min)	5	0.2	-

3.3.1 Key Observations

The comparative analysis between FEA and analytical results showed very close agreement for equivalent stress, with a small error of approximately 2.33%, confirming the reliability of both methods. However, a larger discrepancy of 42.75% was observed in shear stress, likely due to modeling simplifications in the analytical approach. In terms of computational efficiency, FEA required approximately 5 minutes to solve, while the analytical method provided results in under 15 seconds. Design verification confirmed that stresses remained well below material limits, ensuring a comfortable safety margin. Additionally, the critical stress locations identified by FEA corresponded closely with those predicted analytically, reinforcing the consistency and reliability of both approaches for structural assessment.

3.3.2 Methodology Comparison

Table 3.3: Advantages and Limitations of Each Method

Finite Element Analysis (FEA)	Analytical Method
Strengths:	Strengths:
- Detailed stress distribution visualization - Capable of handling complex geometries - Ability to detect local stress concentrations	- Very fast and easy calculations - Direct compliance with design codes - Minimal computational resources
Limitations:	Limitations:
- Requires careful mesh convergence study - High computational cost for fine meshes - Needs validation against theory/experiments	- Conservative simplifications - No insight into local peak stresses - Less suitable for irregular geometries

3.4 Recommended Workflow and Conclusion

Based on the updated comparative study of the T-joint fillet weld using both Finite Element Analysis (FEA) and analytical calculations, the following workflow is recommended for efficient and reliable structural verification:

First, use analytical methods for preliminary weld sizing and quick feasibility checks. This allows for a rapid assessment and ensures initial compliance with code requirements. Next, perform FEA using a moderate mesh density (around 1.5–5mm element size for similar structures) for detailed analysis. If needed, apply localized mesh refinement around geometric discontinuities such as fillet radii and attachments to capture stress concentrations more accurately. Finally, cross-validate critical stress values between the analytical and numerical models to ensure the accuracy and reliability of the design.

The investigation demonstrated that the design is safe with a significant margin, as all computed stresses remained well below the material allowable limits. FEA revealed local stress concentrations that were not captured by analytical methods, highlighting its importance for critical structures, while analytical methods enabled extremely fast preliminary verification and ensured compliance with EN 1993-1-8.

Based on these findings, the recommended approach is to use analytical methods for quick sizing and code checks, FEA for detailed stress analysis and hotspot detection, and to cross-verify both methods for maximum reliability and structural confidence. The combination of these methods offers a balanced approach, supporting both fast initial checks and more detailed, comprehensive analysis.

Overall, this study strongly supports the use of a combined analytical and numerical approach for achieving safe, efficient, and code-compliant structural design verification.

Chapter 4 References

- ANSYS Workbench 2023 R1 Documentation
- EN 1993-1-8: Eurocode 3: Design of steel structures - Part 1-8: Design of joints
- Step-by-Step Guide provided during the internship
- Cook, R.D. et al. (2002). Concepts and Applications of Finite Element Analysis
- Megson, T.H.G. (2014). Structural and Stress Analysis